Physical Chemistry (I)

Lecture 22

Electrochemistry

Statistical Mechanics

In the Last Class

- Chemical equilibrium
  - Reaction quotient and equilibrium constant
  - Activity
- Le Chatelier's principle
  - Amount change
  - Temperature change (van't Hoff equation)

Q and K

$$\Delta_\mathrm{r}G = \Delta_\mathrm{r}G^\ominus + RT\ln Q$$

where the reaction quotient Q is defined as

$$Q = \frac{\prod_{\mathrm{product}\ j} a_j^{\nu_j}}{\prod_{\mathrm{reactant}\ i} a_i^{\nu_i}}$$

At equilibrium, $\Delta_\mathrm{r}G = 0$

$$0 = \Delta_\mathrm{r}G^\ominus + RT\ln Q_\mathrm{eq} = \Delta_\mathrm{r}G^\ominus + RT\ln K$$

where the equilibrium constant K is Q at equilibrium

Approximation for Activity

State	Measure	Approximation for a_J	Definition
Solute	molality	$b_J/b^\ominus$	$b^\ominus = 1\,\text{mol}\,\text{kg}^{-1}$
	molar concentration	$[J]/c^\ominus$	$c^\ominus = 1\,\text{mol}\,\text{dm}^{-3}$
Gas phase	partial pressure	$p_J/p^\ominus$	$p^\ominus = 1\,\text{bar}$
Pure solid, liquid		1 (exact)	

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., p. 208)

Last class

Le Chatelier's Principle

When subjected to a disturbance,

a system at equilibrium tends to respond in a way

that minimizes the effect of the disturbance.

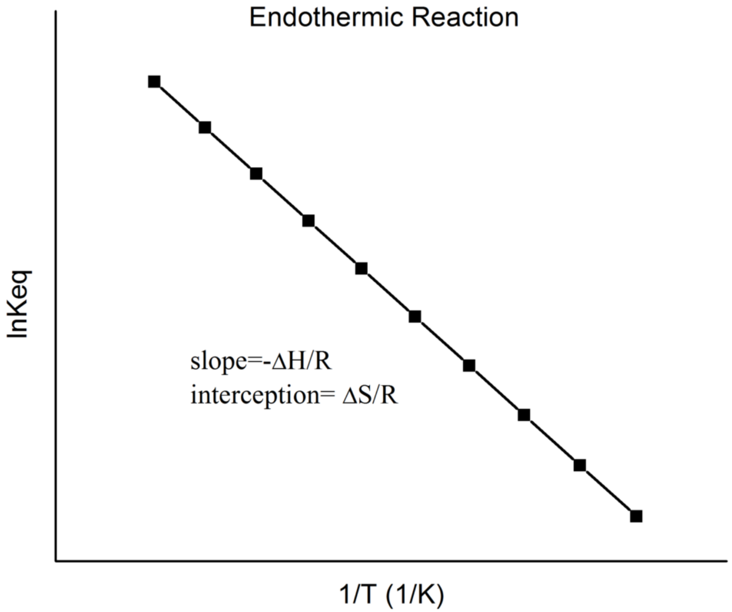
Disturbances to consider:

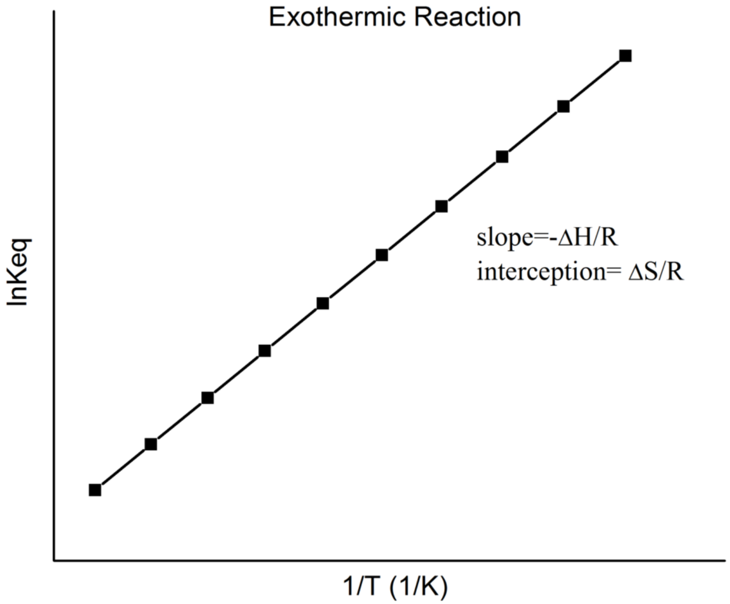
- Amount (concentration, pressure)
- Temperature

Last class

van’t Hoff Equation

$$\frac{\mathrm{d}\ln K}{\mathrm{d}T} = \frac{\Delta_r H^\ominus}{RT^2}$$





(Image: https://en.wikipedia.org/wiki/Van_%27t_Hoff_equation)

Thermodynamics of Cell

Recall that

$$\mathrm{d}G_{T,p} = \delta w_{\mathrm{add,max}} = \delta w_{\mathrm{elec}}$$

for a reversible electrochemical reaction.

For an infinitesimal progress of the reaction $\mathrm{d}\xi$,

$$\mathrm{d}G = \Delta_{\mathrm{r}} G \mathrm{d}\xi$$

Hence,

$$\delta w_{\mathrm{elec}} = \Delta_{\mathrm{r}} G \mathrm{d}\xi$$

Thermodynamics of Cell

$$\delta w_{\mathrm{elec}} = \Delta_{\mathrm{r}} G \mathrm{d}\xi$$

On the other hand, the electrical work is

$$\delta w_{\mathrm{elec}} = \phi \mathrm{d}Q$$

where ϕ is the electric field and dQ is the transferred charge. For the electrochemical reaction,

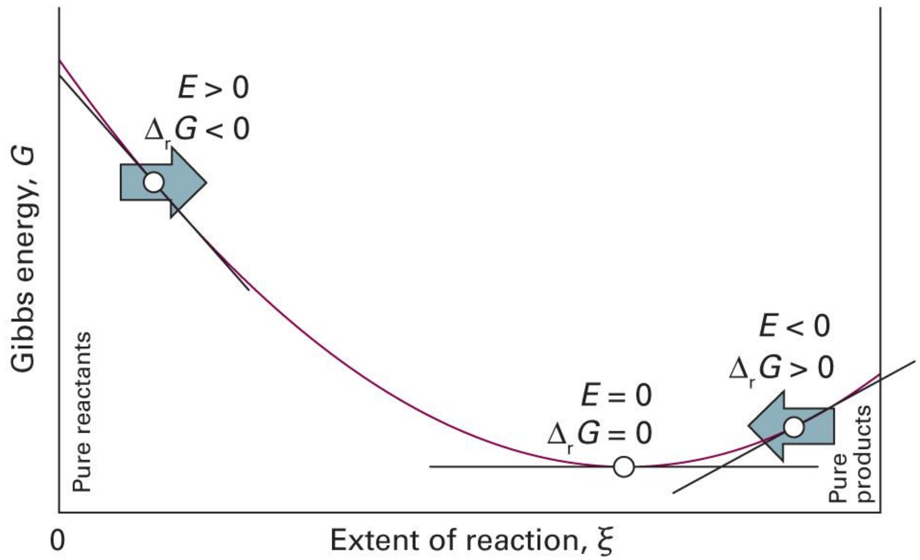
$$\phi = E_{\mathrm{cell}},\ \mathrm{d}Q = -\nu e N_{\mathrm{A}} \mathrm{d}\xi = -\nu F \mathrm{d}\xi$$

$$\delta w_{\mathrm{elec}} = -\nu F E_{\mathrm{cell}} \mathrm{d}\xi$$

Thus, $\Delta_{\mathrm{r}} G = -\nu F E_{\mathrm{cell}}$

Cell Potential and Reaction

$$\Delta_r G = -\nu F E_{cell}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 6C.4)

Nernst Equation

$$\Delta_r G = -\nu F E_{cell}$$

Using $\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q$,

$$-\nu F E_{cell} = \Delta_r G^{\ominus} + RT \ln Q$$

Thus,

$$E_{cell} = E^{\ominus}_{cell} - \frac{RT}{\nu F} \ln Q$$

where $E^{\ominus}_{cell} = -\Delta_r G^{\ominus}/\nu F$

At Equilibrium

$$E_{\text{cell}} = E^{\ominus}_{\text{cell}} - \frac{RT}{\nu F}\ln Q$$

At equilibrium, $E_{\text{cell}} = 0$ and $Q = K$

$$0 = E^{\ominus}_{\text{cell}} - \frac{RT}{\nu F}\ln K$$

or,

$$E^{\ominus}_{\text{cell}} = \frac{RT}{\nu F}\ln K$$

Topic 6C

Temperature Coefficient

$$E^{\ominus}_{\text{cell}} = -\frac{\Delta_{\text{r}} G^{\ominus}}{\nu F}$$

Since $(\partial G/\partial T)_p = -S$,

$$\frac{\mathrm{d}E^{\ominus}_{\text{cell}}}{\mathrm{d}T} = -\frac{1}{\nu F}\left(\frac{\partial \Delta_{\text{r}} G^{\ominus}}{\partial T}\right)_p = \frac{\Delta_{\text{r}} S^{\ominus}}{\nu F}$$

Furthermore,

$$\Delta_{\text{r}} H^{\ominus} = \Delta_{\text{r}} G^{\ominus} + T\Delta_{\text{r}} S^{\ominus} = -\nu F\left(E^{\ominus}_{\text{cell}} - T\frac{\mathrm{d}E^{\ominus}_{\text{cell}}}{\mathrm{d}T}\right)$$

Electrode Potentials

Cell potential = cathode contribution + anode contribution

For a cell L‖R,

$$E^{\ominus}_{\text{cell}} = E^{\ominus}(\text{R}) - E^{\ominus}(\text{L})$$

Impossible to measure each contribution

Convention

- Standard hydrogen electrode (SHE): $\mathrm{Pt}(s)|\mathrm{H_2}(g)|\mathrm{H^+}(aq)$

$$E^{\ominus} = 0 \text{ at all temperatures}$$

- Standard potential $E^{\ominus}(\mathrm{X})$

For a cell $\mathrm{Pt}(s)|\mathrm{H_2}(g)|\mathrm{H^+}(aq)\|\mathrm{X}$, $E^{\ominus}(\mathrm{X}) = E^{\ominus}_{\mathrm{cell}}$

Standard Potentials

Couple	$E^{\ominus}/\mathrm{V}$
$Ce^{4+}(aq) + e^- \rightarrow Ce^{3+}(aq)$	+1.61
$Cu^{2+}(aq) + 2\,e^- \rightarrow Cu(s)$	+0.34
$AgCl(s) + e^- \rightarrow Ag(s) + Cl^-(aq)$	+0.22
$H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g)$	0
$Zn^{2+}(aq) + 2\,e^- \rightarrow Zn(s)$	−0.76
$Na^+(aq) + e^- \rightarrow Na(s)$	−2.71

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Table 6D.1)

Thermodynamics of Potentials

Each electrode potential can be converted into

$$-\nu F E^{\ominus} = \Delta_r G^{\ominus}$$

When a combination of reactions a and b gives reaction c,

$$\Delta_r G^{\ominus}(\mathrm{c}) = \Delta_r G^{\ominus}(\mathrm{a}) + \Delta_r G^{\ominus}(\mathrm{b})$$

$$-\nu_\mathrm{c} F E^{\ominus}(\mathrm{c}) = -\nu_\mathrm{a} F E^{\ominus}(\mathrm{a}) - \nu_\mathrm{b} F E^{\ominus}(\mathrm{b})$$

Thus,

$$\nu_\mathrm{c} E^{\ominus}(\mathrm{c}) = \nu_\mathrm{a} E^{\ominus}(\mathrm{a}) + \nu_\mathrm{b} E^{\ominus}(\mathrm{b})$$

Reducing Power

For two redox couples Ox_L/Red_L and Ox_R/Red_R,

the cell reaction $Red_L + Ox_R \rightarrow Ox_L + Red_R$ has

- $K > 1$: $E^{\ominus}_{\text{cell}} > 0$ and $E^{\ominus}(\text{L}) < E^{\ominus}(\text{R})$
- $K < 1$: $E^{\ominus}_{\text{cell}} < 0$ and $E^{\ominus}(\text{L}) > E^{\ominus}(\text{R})$

Red_L has a thermodynamic tendency to reduce Ox_R

if $E^{\ominus}(\text{L}) < E^{\ominus}(\text{R})$

Electrochemical Series

Least strongly reducing
Gold (Au^{3+}/Au)
Platinum (Pt^{2+}/Pt)
Silver (Ag^{+}/Ag)
Mercury (Hg^{2+}/Hg)
Copper (Cu^{2+}/Cu)
Hydrogen (H^{+}/H_2)
Tin (Sn^{2+}/Sn)
Nickel (Ni^{2+}/Ni)
Iron (Fe^{2+}/Fe)
Zinc (Zn^{2+}/Zn)
Chromium (Cr^{3+}/Cr)
Aluminium (Al^{3+}/Al)
Magnesium (Mg^{2+}/Mg)
Sodium (Na^{+}/Na)
Calcium (Ca^{2+}/Ca)
Potassium (K^{+}/K)
Most strongly reducing

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Table 6D.2)

Summary

$$dG_{T,p} = \delta w_{\text{add,max}}$$

Electrochemical cell

- Cell potential

$$\Delta_r G = -\nu F E_{\text{cell}}$$

- Nernst equation

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln Q$$

- Electrode potentials

10. Statistical Mechanics

Chapter Overview

- Maxwell-Boltzmann statistics
  - Boltzmann factor and partition function
  - Classical statistical mechanics: phase space
- Partition function and thermodynamics
  - Generalization of partition function
  - Partition function and thermodynamic quantities
- Applications
  - Perfect gas and real gas
  - Equipartition theorem

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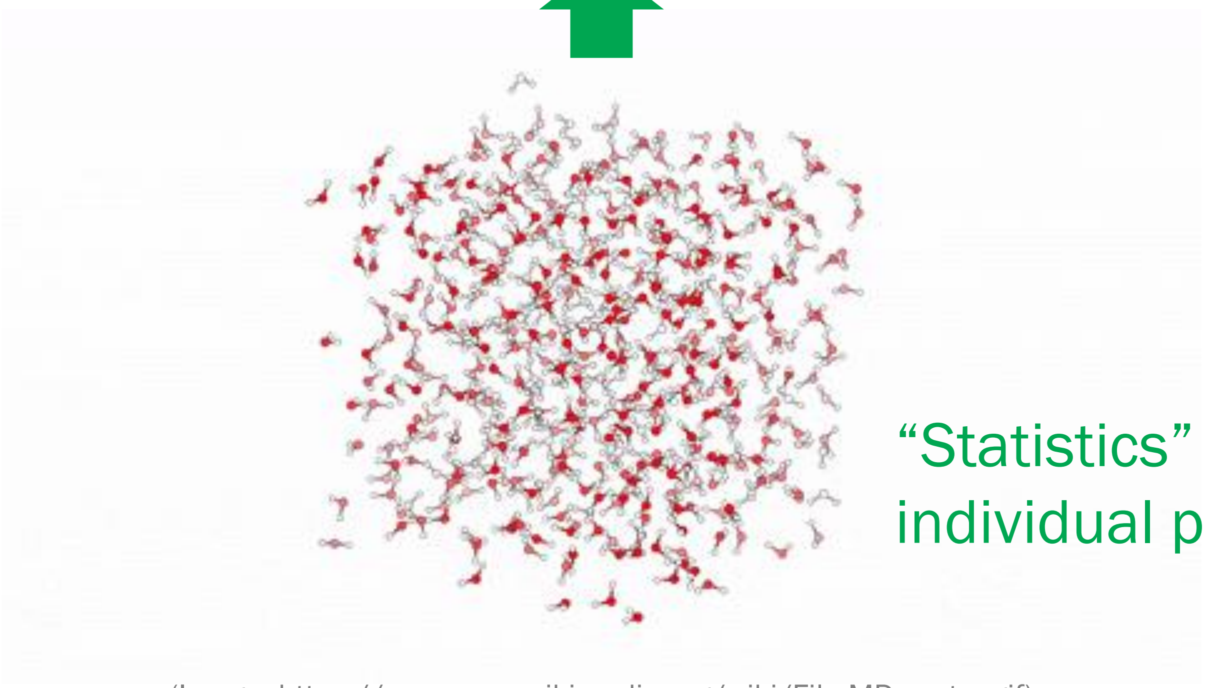
Statistical Mechanics

Thermodynamics (p, V, T, ...)



“Statistics” of individual particles

(Image: https://commons.wikimedia.org/wiki/File:MD_water.gif)

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Maxwell-Boltzmann Statistics

Average distribution of independent particles

of a system in thermal equilibrium

Described by the Boltzmann factor

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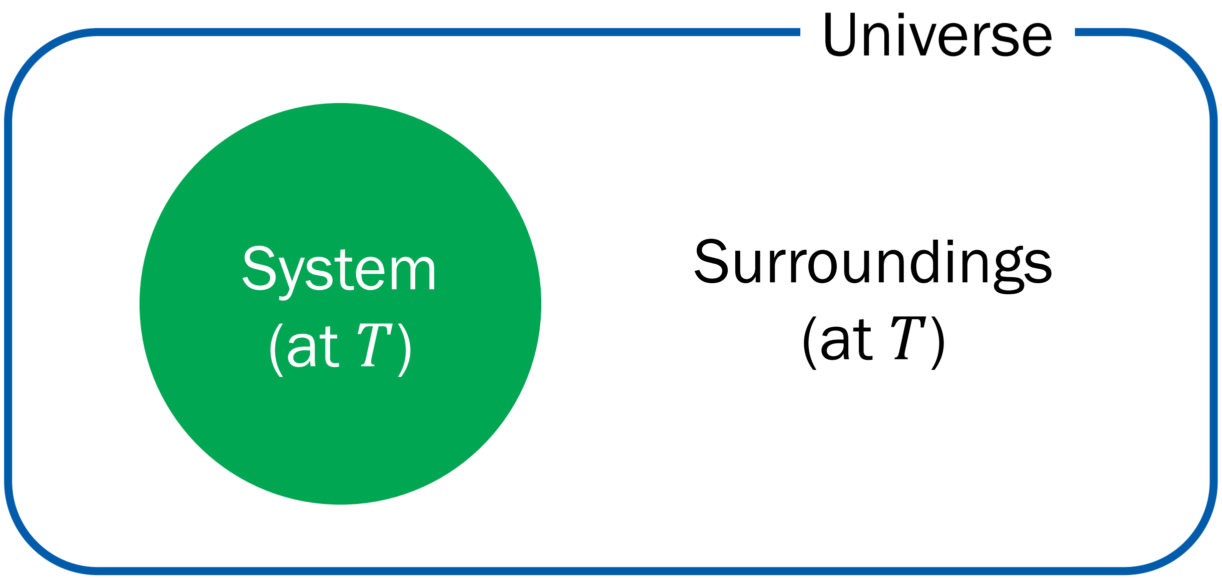
System Microstates

For a closed system with constant V and n,



Consider two microstates s_1 and s_2.

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Probabilities

- Probability of having s_i is proportional to the number of microstates in the surroundings

$$P(s_i) \propto W_{\text{surr}}(s_i)$$

- Ratio between probabilities of having s_1 and s_2

$$\frac{P(s_2)}{P(s_1)} = \frac{W_{\text{surr}}(s_2)}{W_{\text{surr}}(s_1)}$$

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Entropy Comes in

$$\frac{P(s_2)}{P(s_1)} = \frac{W_{\text{surr}}(s_2)}{W_{\text{surr}}(s_1)}$$

Using the statistical definition of entropy, $S = k_{\text{B}} \ln W$,

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{S_{\text{surr}}(s_2)/k_{\text{B}}}}{e^{S_{\text{surr}}(s_1)/k_{\text{B}}}} = e^{\{S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)\}/k_{\text{B}}}$$

What is $S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)$?

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Entropy of Surroundings

Recall the fundamental equation of thermodynamics:

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n = T\mathrm{d}S \text{ (at constant } V \text{ and } n\text{)}$$

For constant T, $\Delta U = T\Delta S$ and

$$S_{\mathrm{surr}}(s_2) - S_{\mathrm{surr}}(s_1) = \frac{1}{T}\{U_{\mathrm{surr}}(s_2) - U_{\mathrm{surr}}(s_1)\}$$

$$= -\frac{1}{T}\{U(s_2) - U(s_1)\}$$

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Written in System Variables

$$\frac{P(s_2)}{P(s_1)} = e^{\{S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)\}/k_{\text{B}}}$$

$$S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1) = -\frac{1}{T}\{U(s_2) - U(s_1)\}$$

Combining the two equations,

$$\frac{P(s_2)}{P(s_1)} = e^{-\{U(s_2) - U(s_1)\}/k_{\text{B}}T} = \frac{e^{-U(s_2)/k_{\text{B}}T}}{e^{-U(s_1)/k_{\text{B}}T}}$$

Boltzmann Factor

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{-U(s_2)/k_\mathrm{B}T}}{e^{-U(s_1)/k_\mathrm{B}T}}$$

In general, for a microstate s with energy E,

$$P(s) \propto e^{-E(s)/k_\mathrm{B}T} \text{ (Boltzmann factor)}$$

or, using a constant q,

$$P(s) = \frac{1}{q} e^{-E(s)/k_\mathrm{B}T}$$

Determination of q

$$P(s) = \frac{1}{q} e^{-E(s)/k_{\mathrm{B}}T}$$

Using the normalization condition, $\sum_s P(s) = 1$,

$$1 = \sum_s \frac{1}{q} e^{-E(s)/k_{\mathrm{B}}T} = \frac{1}{q} \sum_s e^{-E(s)/k_{\mathrm{B}}T}$$

$$q = \sum_s e^{-E(s)/k_{\mathrm{B}}T}$$

(Partition function)

Brief Illustration 12A.2

Suppose that two conformations of a molecule differ in energy by 5.0 kJ mol^{-1} (corresponding to 8.3×10^{-21} J for a single molecule), so conformation A lies at energy 0 and conformation B lies at ε = 8.3×10^{-21} J. What is the proportion of molecules in conformation B at 293 K?

$E(\mathrm{A})/k_{\mathrm{B}}T$ = 0/{(1.381×10^{-23} J K^{-1})$\times$(293 K)} = 0

$E(\mathrm{B})/k_{\mathrm{B}}T$ = (8.3×10^{-21} J)/{(1.381×10^{-23} J K^{-1})$\times$(293 K)} = 2.05

The partition function is

$$q=\sum_s e^{-E(s)/k_{\mathrm{B}}T}=e^{-E(\mathrm{A})/k_{\mathrm{B}}T}+e^{-E(\mathrm{B})/k_{\mathrm{B}}T}=e^{0}+e^{-2.05}=1.13$$

Brief Illustration 12A.2

$$q = 1.13$$

Hence, the proportion of molecules in conformation B is

$$P(\mathrm{B}) = \frac{1}{q} e^{-E(\mathrm{B})/k_{\mathrm{B}}T} = \frac{1}{1.13} e^{-2.05} = 0.11$$

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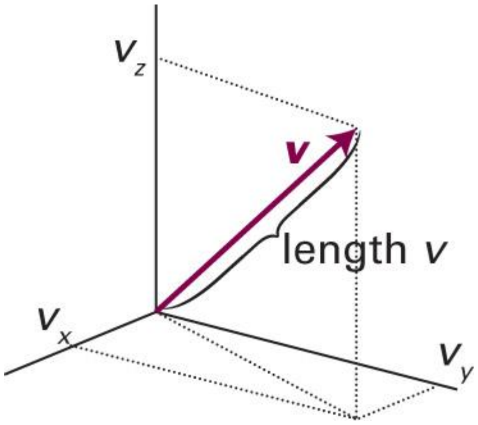
Classical Statistical Mechanics

What is a microstate of a classical particle?

Position and Velocity



Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \left(\frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}, \frac{\mathrm{d}z}{\mathrm{d}t}\right)$$

With the velocity, we can define:

- Speed $v = |\vec{v}| = \left(v_x^2 + v_y^2 + v_z^2\right)^{1/2}$
- Kinetic energy $E_\mathrm{k} = mv^2/2 = m\left(v_x^2 + v_y^2 + v_z^2\right)/2$

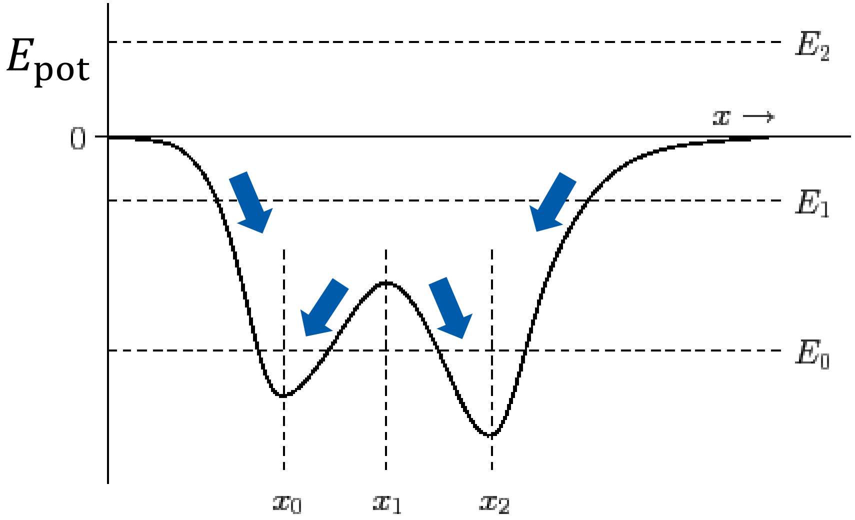
(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 3, Sketch 1)

Review

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Force $\vec{F}$: negative slope of potential energy ($-\nabla E_{\text{pot}}$)



(Image: http://farside.ph.utexas.edu/teaching/336k/Newtonhtml/node16.html)

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Acceleration $\vec{a}$: second time-derivative of position

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

$$-\nabla E_{\text{pot}} = m\frac{d^2\vec{r}}{dt^2}$$

given

Second-Order ODE

$$\frac{d^2\vec{r}}{dt^2} = -\frac{\nabla E_{\text{pot}}}{m}$$

Two conditions are required to obtain the full dynamics:

(initial) position and (initial) velocity (= momentum)

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Phase Space

Space of all possible microstates of a system

- Classical mechanics:

Space of all possible values of

position and momentum variables

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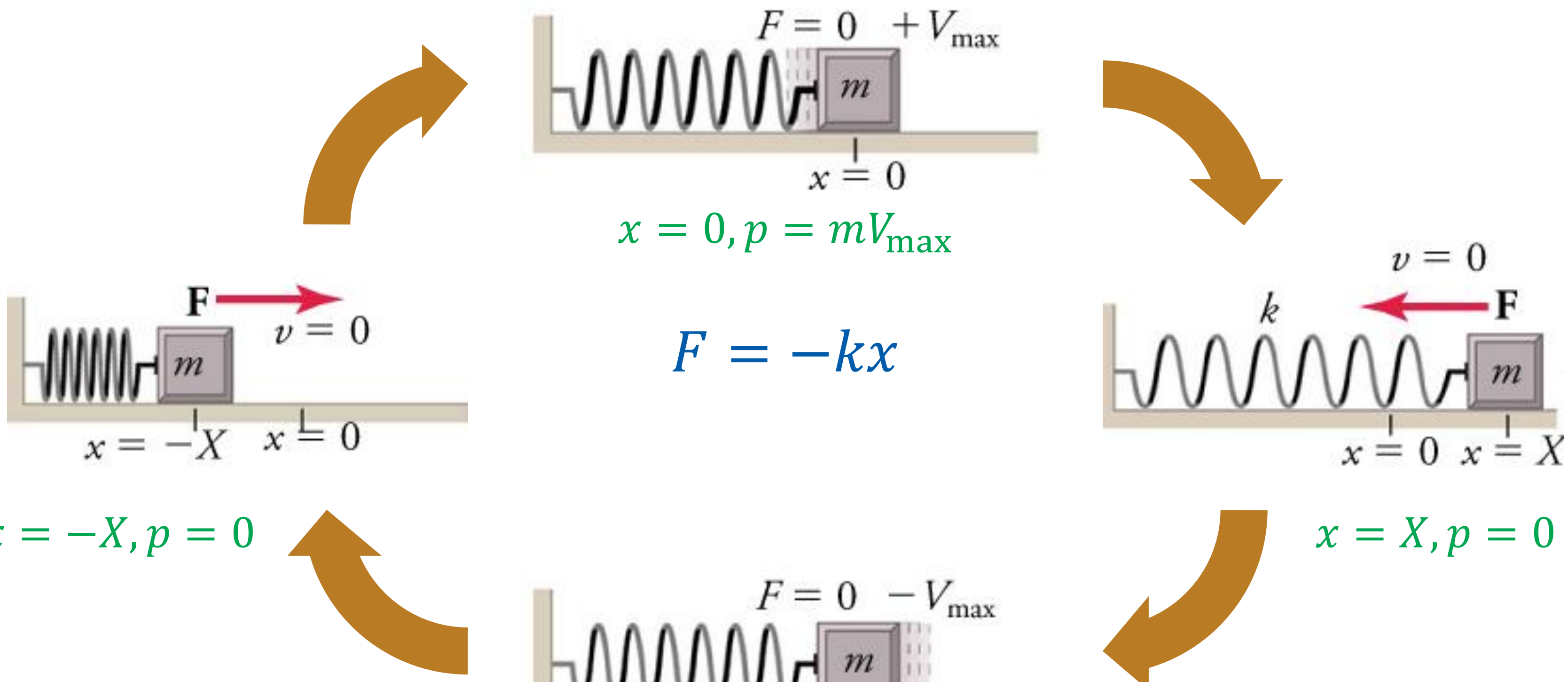
1-Dimensional Spring



$x = 0, p = mV_{\max}$

$$F = -kx$$

$x = X, p = 0$

$x = 0, p = -mV_{\max}$

$x = -X, p = 0$

(Image: https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion)

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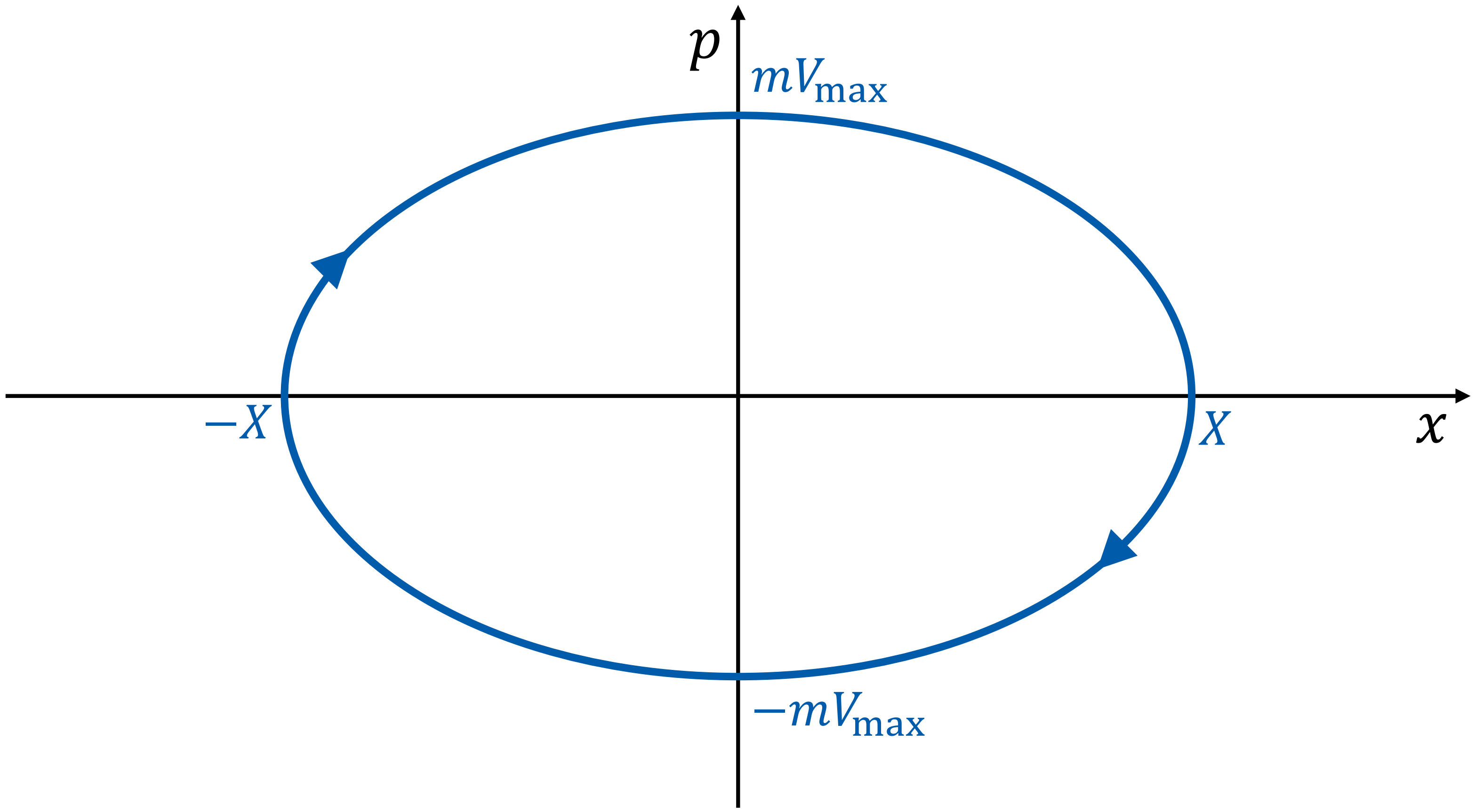
Phase Space Representation



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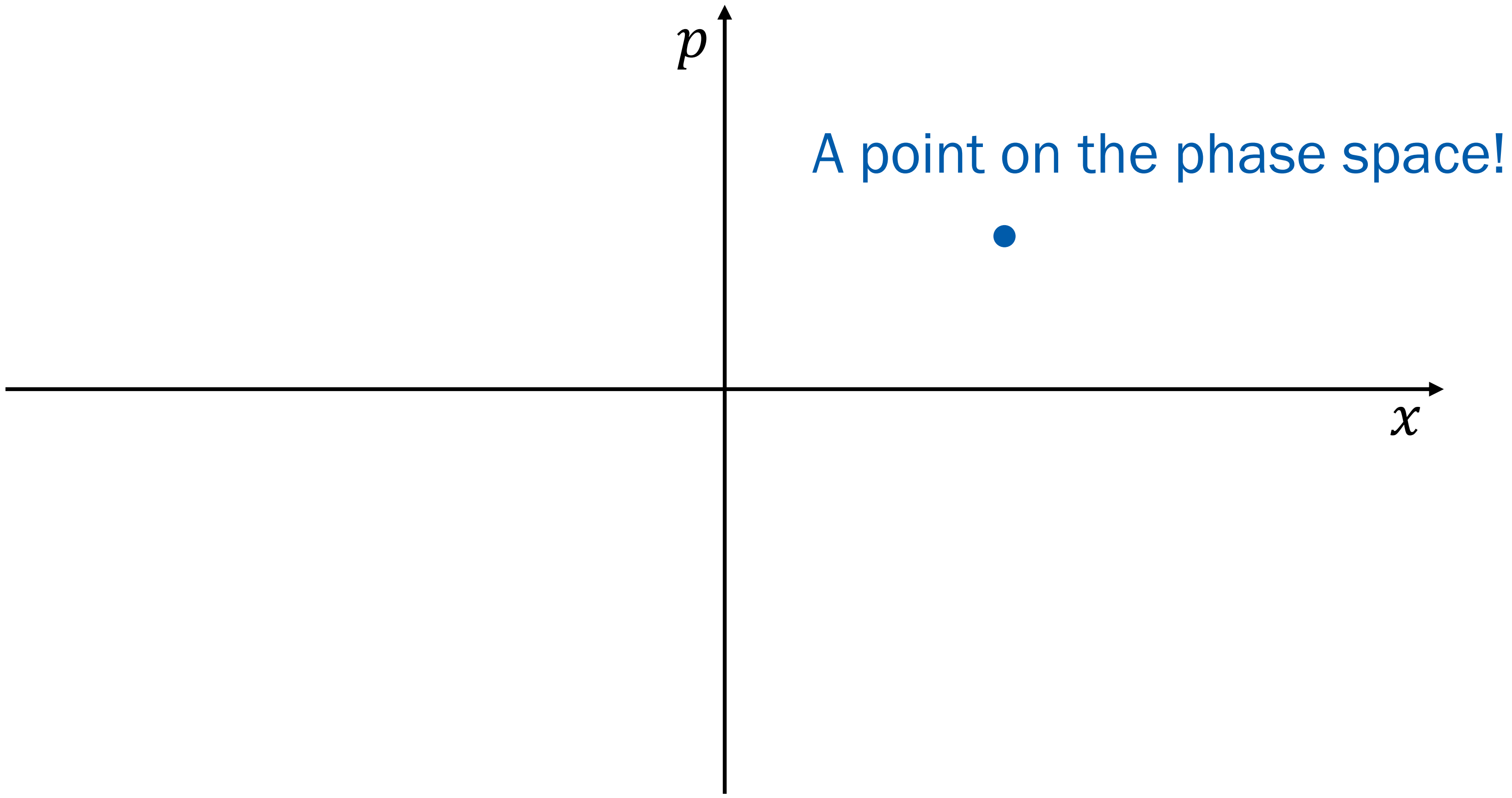
Microstate of Classical Particle



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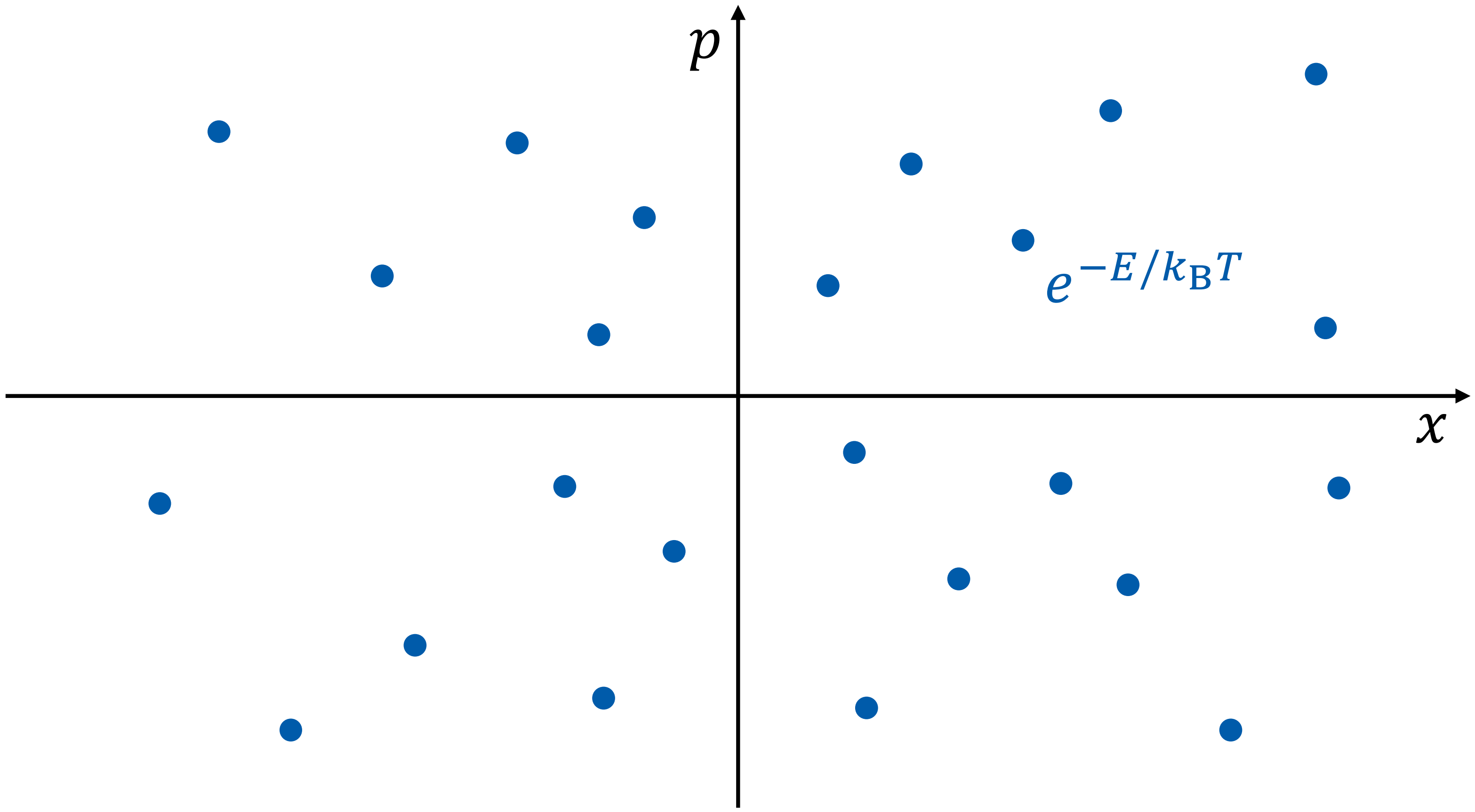
Partition Function



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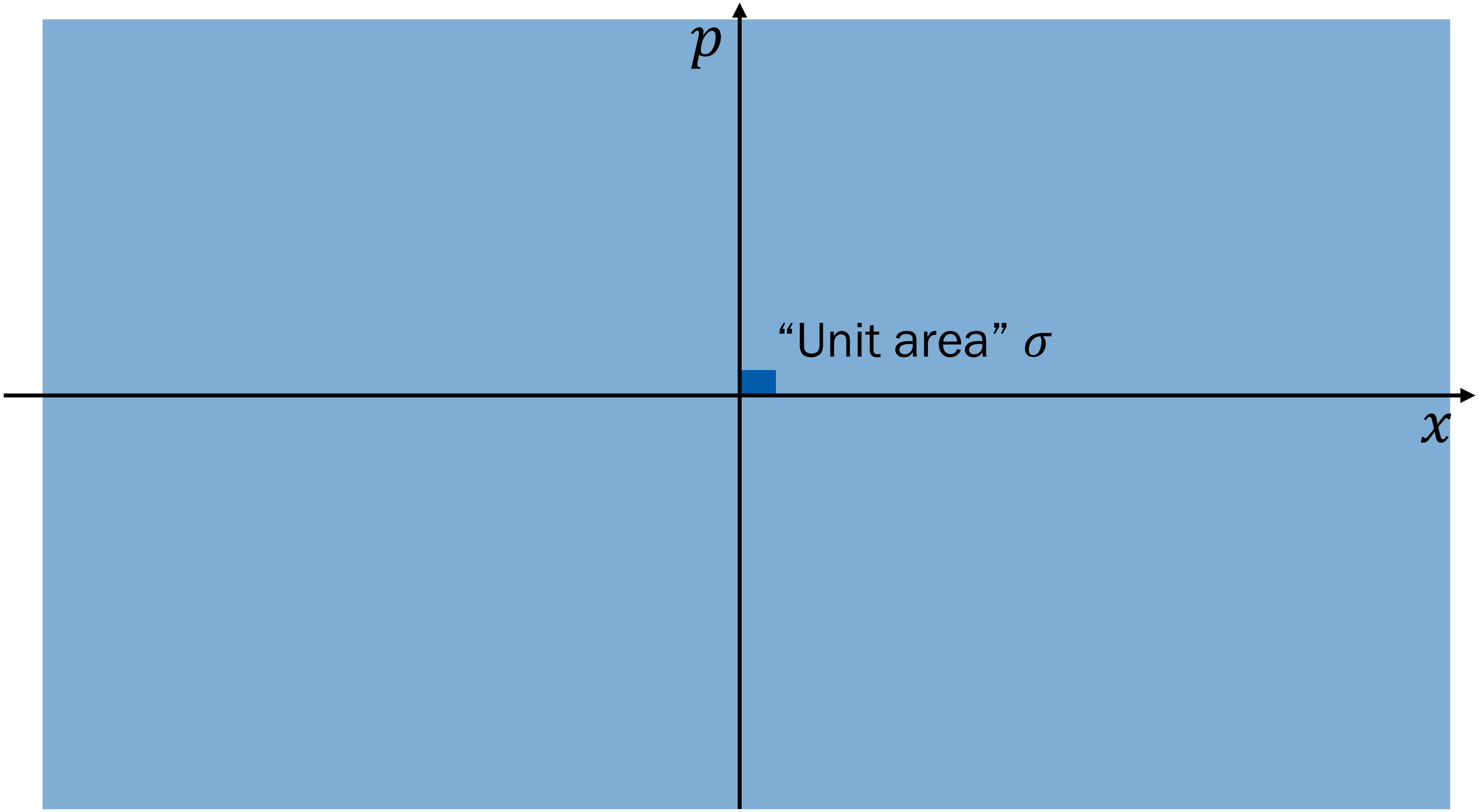
Partition Function



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Partition Function

$$q = \sum_{s} e^{-E(s)/k_{\mathrm{B}}T} \;\rightarrow\; q = \frac{1}{\sigma}\int\int e^{-E(x,p)/k_{\mathrm{B}}T}\,dxdp$$

for a 1-dimensional single particle

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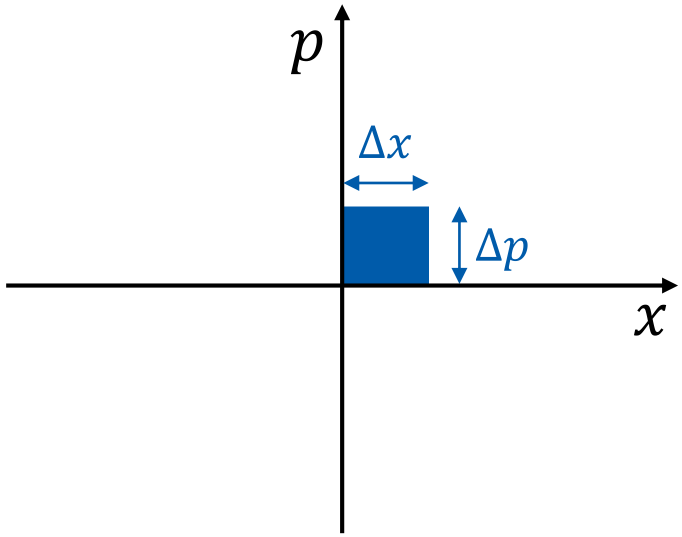
What is σ?

From Heisenberg's uncertainty principle,

$\Delta x \Delta p \geq h$ (Planck's constant)



Take $\sigma = h$!

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Partition Function

$$q = \frac{1}{h}\int\int e^{-E(x,p)/k_\mathrm{B}T}\,dxdp$$

- Example: 1-dimensional spring

$$E(x,p) = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kx^2 + \frac{p^2}{2m}$$

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{1}{k_\mathrm{B}T}\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,dxdp$$

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1-Dimensional Spring

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{1}{k_{\mathrm{B}}T}\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,dxdp$$

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}-\frac{p^2}{2mk_{\mathrm{B}}T}}\,dxdp$$

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}} \times e^{-\frac{p^2}{2mk_{\mathrm{B}}T}}\,dxdp$$

$$q = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}}\,dx\right]\left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_{\mathrm{B}}T}}\,dp\right]$$

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Gaussian Integrals

$$q = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_\mathrm{B}T}}\, dx\right]\left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_\mathrm{B}T}}\, dp\right]$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2}\, dx = \sqrt{\pi/\alpha}$,

$$q = \frac{1}{h}\sqrt{\frac{2\pi k_\mathrm{B}T}{k}}\sqrt{2m\pi k_\mathrm{B}T} = \frac{2\pi k_\mathrm{B}T}{h}\sqrt{\frac{m}{k}}$$